# Computer Engineering Technology (TEJ3M) Course Outline

Port Credit Secondary School, Peel District School Board

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| Grade, Course Type & Subject: Grade 11, University/College Preparation, Computer Engineering Technology | |
| Credit Value: 1.0 | Prerequisite(s): None |
| Course Teacher: Andrew Andrade<br>Reviewed by: Shaun Knowles | Teacher Email: andrew.andrade@peelsb.com |
| Department: Technology & Computer Science | Department Head: Shaun Knowles |
| Ministry Curriculum/Program Document(s): The Ontario Curriculum, Grades 11 and 12: Technological Education, 2009 (revised), Computer Engineering Technology (TEJ3M). | |

## COURSE CONTENT

Course Description: This Grade 11 university/college preparation course examines computer systems and the control of external devices. Students develop knowledge and skills in computer hardware and operating systems, electronics, programming, networking, robotics, automation, and interfacing. Course work emphasizes safe practice, systematic troubleshooting, technical communication, environmental and societal considerations, and pathways related to computer engineering technology.

**Overarching Learning Goals:**

- *Work safely and responsibly in a respectful, positive, inclusive, and identity-affirming technological learning environment that supports belonging and well-being.*
- *Demonstrate professional work habits through empathy, collaboration, leadership, communication, documentation, and care of tools, equipment, materials, and shared spaces.*
- *Apply computer engineering concepts to build, configure, test, and troubleshoot course-related hardware, software, electronic, network, and control systems.*
- *Use systematic problem solving, programming, measurement, and technical documentation to develop and improve computer engineering solutions.*
- *Exercise agency and ownership in learning, consider environmental and societal impacts, and connect computer engineering technology to real-world applications, community contexts, and multiple postsecondary and career pathways.*

Overall Curriculum Expectations: TEJ3M expectations are organized under Computer Technology Fundamentals; Computer Technology Skills; Technology, the Environment, and Society; and Professional Practice and Career Opportunities.

| Content Focus Areas | Learning Goals |
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| *Safety* | Apply safe procedures, hazard awareness, PPE, and appropriate use of tools and equipment while contributing to a safe, respectful, and positive learning environment. |
| Professionalism | Demonstrate preparation, workmanship, empathy, collaboration, leadership, communication, documentation, organization, environmental stewardship, and responsible use of resources. |
| Technical Knowledge and Troubleshooting | Apply computer engineering concepts, measurement, logic, and systematic troubleshooting while building, configuring, testing, and improving course-related systems. |
| Systems, Programming and Control | Apply hardware, operating-system, networking, electronics, programming, interfacing, and control concepts in course-related technical work. |
| Documentation, Impact and Pathways | Document and communicate technical work professionally, consider environmental and societal impacts, and connect learning to real-world applications, community contexts, and education and career pathways related to computer engineering technology. |

### Resources

- *Course text(s): When needed, course materials are provided digitally through Google Classroom and on [andrewandrade.ca](andrewandrade.ca). The website is public and provides information only; it does not collect user data or personal information. It presents course content in a clearer, more organized format than Google Classroom and links to view-only Google Drive materials such as slides and worksheets. It also includes separate course notes that supplement and enhance the textbooks used in the course.*
- *VLE: Google Classroom*
- *Calculator: Casio fx-991ES PLUS 2nd Edition is recommended.*
- *Materials: Core materials required to meet curriculum expectations will be provided. Optional additional or enhanced materials may be offered where approved. Lost, unreturned, or damaged school property may be subject to repair or replacement procedures under current school and board policy.*
- *Students may bring their own tools, materials, and equipment with teacher permission and only where permitted by school and board policies. Personal items must be approved before use.*
- *Digital services: This course outline, including Appendix A, provides notice of the anticipated third-party software and digital services that may be used in the course. Students should use their PDSB school account where an account is required and follow the privacy expectations in Appendix A. Required submission formats still apply regardless of the software used.*

## TEACHING STRATEGIES TO SUPPORT STUDENT LEARNING

Teaching and learning are grounded in Universal Design for Learning and Culturally Responsive and Relevant Pedagogy. Students may learn through demonstrations, hands-on practice, projects, examples, discussion, collaboration, technical documentation, digital tools, feedback, and reflection. Instruction may be adjusted to support student readiness, safety, accessibility, available equipment and materials, and authentic learning opportunities while maintaining the curriculum expectations.

Homework: Work outside class may be used for practice, consolidation, preparation, or appropriate portions of longer tasks. Practical skills that require specialized equipment, supervision, or teacher observation must be demonstrated in class.

## STUDENT RESPONSIBILITIES

As active agents in their own learning, it is expected that students will:

- Consistently engage in the learning process.
- Seek clarification when unsure of expectations, learning goals, or success criteria.
- Apply feedback to improve their learning.
- Engage in opportunities to set individual goals, self-assess, and reflect on their progress.
- Provide authentic evidence of their learning within the agreed-upon time frame.
- Ensure that teachers have multiple sources of evidence to accurately determine their level of achievement.

## PROFESSIONAL SUBMISSION STANDARDS

Unless otherwise directed, formal written reports and similar submissions should be submitted as PDF files and identified with the student's number. Software-based work may use the format, file, repository, link, or export specified for the task. When a task requires both an editable or shareable artifact and a fixed record, both must be submitted. For example, a presentation may require the Google Slides link and a PDF version of the presentation.

## PRACTICAL WORK CHECKPOINTS

Safety Checkpoint: Complete required safety instruction and demonstrate safe procedures, PPE, and authorized use of equipment.
Theory Checkpoint: Demonstrate the prerequisite knowledge, planning, or calculations needed for the task.
Responsibility Checkpoint: Demonstrate focused, responsible conduct, appropriate use and return of tools or materials, and care of the workspace.

Practical access to tools, materials, and equipment may be paused when a required checkpoint has not yet been met and restored through review, retraining, reassessment, or another appropriate process. Practical activities are scheduled in batches because machines, tools, equipment, setup, maintenance, and supervision are shared. A student who is not ready during the scheduled practical window may lose that specific hands-on opportunity for logistical reasons. Work may still be submitted and assessed using available curriculum-aligned evidence. Theory, simulation, design, documentation, observation, or other alternative work may be assigned when practical access is not available or appropriate. Where a curriculum expectation requires practical demonstration, sufficient evidence of that expectation is still required.

## SCHEDULED SAFETY DEMONSTRATIONS

Safety demonstrations, training, and authorization checks will be scheduled with students and announced in class. Students are expected to attend and complete the required demonstration within the scheduled training period. Multiple opportunities will normally be provided, but these opportunities are time-limited because equipment, machines, setup, maintenance, and supervision must be scheduled for the class.

Students who are absent, arrive late, are not prepared, or postpone completing a required safety demonstration may have fewer remaining opportunities. Once the announced training period has ended, an additional demonstration may not be available simply because a student is ready later. A student who has not completed the required safety demonstration may still complete and submit appropriate theory or other curriculum-aligned work, but may not be permitted to participate in the related hands-on activity.

## AI AND AUTHENTIC WORK

AI may be used as a learning tool unless a task says otherwise. Use it to support thinking, not replace it. Students must think critically, verify important information, make their own decisions, and write final explanations in their own words. Do not copy and submit AI-generated work as your own. Students may be asked to provide prompts, drafts, version history, or explain what they kept, changed, rejected, tested, or verified. Personal, identifying, confidential, or restricted school information must not be entered into unapproved AI tools.

## ASSESSMENT AND EVALUATION GUIDELINES

**"The primary purpose of assessment and evaluation is to improve student learning."**
**- Ministry of Education, Growing Success**

Assessment, evaluation, and reporting follow Growing Success: Assessment, Evaluation and Reporting, K-12 (2026), current PDSB Assessment, Evaluation, and Reporting policy and procedures, and applicable school and department direction.
Assessment and evaluation are based on the provincial curriculum expectations. Students will have varied opportunities to practise, receive feedback, and demonstrate learning. Evidence may be gathered through observations, conversations, and products, including practical demonstrations, projects, technical documentation, tests or quizzes, designs, code, troubleshooting, presentations, and other appropriate evidence.
Final grades reflect achievement of the overall curriculum expectations, using the most consistent evidence with special consideration given to more recent evidence.

For Technological Education courses in 2026-27, the final grade is determined as follows:

- Grades 9-10: 65% classroom work, 10% written final exam, 10% culminating task, and 15% attendance and participation.
- Grades 11-12: 65% classroom work, 10% written final exam, 15% culminating task, and 10% attendance and participation. Grades 11-12 workplace preparation and open courses are exempt from the mandatory written exam requirement.

| **Final Evaluation** |
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| Written final exam (10%) and culminating computer engineering technology project or task (15%). Excused absences do not negatively affect the attendance component.<br><br>Late and missed work: Students are responsible for providing evidence of learning within communicated timelines. Any mark consequence, alternate opportunity, or other response will follow the Ministry, PDSB, school, and department procedures in effect at the time. Local implementation details may be updated as guidance is finalized. |

Learning Skills and Work Habits: Responsibility, organization, independent work, collaboration, initiative, and self-regulation are evaluated separately from curriculum achievement, except where the evidence also demonstrates an explicit curriculum expectation.

Professional practices such as safety, workmanship, technical communication, documentation, and appropriate use of tools or equipment may contribute to the course mark when they are directly connected to curriculum expectations and communicated success criteria.

| Learning Skill | Student Expectations |
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| Responsibility & Organization: | Follow safety procedures, care for tools and equipment, maintain organized physical and digital work, meet communicated timelines, and submit authentic work in the required format. |
| Independent Work, Attendance & Engagement: | Use class time productively, begin work when expectations are clear, use available resources, ask focused questions, and catch up on missed learning. Attendance and participation are evaluated using current Ministry and PDSB criteria; excused absences do not reduce this component. |
| Collaboration & Academic Honesty: | Contribute fairly to shared work, submit authentic evidence of individual learning, and follow the AI and academic integrity expectations in this outline. |
| Self-Regulation: | Manage attention, device use, behaviour, and pace so learning and practical work remain safe and productive. Stop and ask for clarification when unsure, especially when safety is involved, and use feedback to improve. |
| Initiative: | Ask useful questions, seek feedback or support when needed, attempt reasonable troubleshooting, and pursue approved enhancement work after core expectations are complete. |

PCSS is committed to creating an environment for students, staff, and families that supports overall wellness and respects boundaries between work and personal time:

- Teachers will strive to return all communications in a timely fashion.
- Students will do their best to communicate with teachers during school hours. They should plan to speak with teachers during class time if they think they will need support in completing work outside of school.
- Caregivers should be aware that teachers may not immediately respond to communications sent outside of work hours. Teachers will endeavour to respond promptly at available times during the work day.

## APPENDIX A: DIGITAL TOOLS AND PRIVACY NOTICE

This appendix is intended to provide written notice of the anticipated third-party software applications and digital services that may be used in this course, including notice required under Ontario Regulation 52/26 where student personal digital information will be disclosed to a third-party owner or operator.

### Information that may be disclosed

Only the minimum information reasonably required for the educational activity should be used. Depending on the service, this may include a student's PDSB school email or account identifier, student number or initials, course or class information, student-created academic or technical work, files submitted to the service, and service activity associated with the assigned task.

## Purpose

Digital tools may be used for instruction, assessment, feedback, collaboration, technical documentation, computer-aided design, programming and version control, electronics and simulation, data analysis, media creation, portfolios, and other curriculum-related work.

## Legal authority and privacy rights

Disclosure of student personal digital information for educational purposes is subject to the Education Act, the Municipal Freedom of Information and Protection of Privacy Act (MFIPPA), Ontario Regulation 52/26, and applicable PDSB policies and procedures. A parent or guardian of a student under 16, or a student who is 16 or 17, who believes personal digital information has been improperly collected, used, retained, or disclosed may raise the concern with PDSB and exercise the privacy rights available under MFIPPA.

Questions about course use of a digital tool may be directed to Andrew Andrade at andrew.andrade@peelsb.com

## Anticipated third-party or account-based services

- Google Classroom, Google Workspace, Google Colab, and Gemini where approved: Google LLC
- GitHub and GitHub Pages: GitHub, Inc. owned by Microsoft Corporation
- Microsoft 365, MakeCode, and Copilot where approved: Microsoft Corporation
- Onshape: PTC Inc.
- Autodesk services, including Inventor, AutoCAD, Fusion, and Tinkercad: Autodesk, Inc.
- Kaggle: Kaggle Inc.
- SketchUp and related Trimble services: Trimble Inc.
- Canva: Canva Pty Ltd
- Wokwi: Wokwi B.V.

## Other software that may be used

Locally installed, browser-based, or open-source software may include Jupyter, Visual Studio Code or another suitable IDE, RStudio, Arduino IDE, Git, Linux, Node-RED, FreeCAD, LibreCAD, KiCad, Falstad, Fritzing, Photopea, GIMP, Inkscape, Krita, Pencil2D, Wick Editor, OpenToonz, Blender, Audacity, OBS Studio, Shotcut, DaVinci Resolve, and other appropriate tools. Students should not provide personal information to these tools unless specifically directed and the use has been approved.

## Alternative software

The teacher generally recommends board-provided, free, or open-source tools where practical. Students may use another suitable local or offline tool with teacher permission if it can produce the required evidence of learning. Students must tell the teacher what tool they are using so the work can be accessed and assessed appropriately. A different third-party or cloud service that receives student personal digital information must be identified to the teacher before use and may require review or additional notice.

### Student privacy expectations

Students should use the PDSB school account or email provided for school use when an account is required, rather than a personal account. The school account is intended to limit identifying information to school identifiers such as the student number and initials rather than a personal email identity. Students must not add or enter unnecessary personal or sensitive information into third-party services. This includes full names, personal email addresses, home addresses, personal phone numbers, birthdates, health or IEP information, passwords, private personal experiences, or identifying information about another student. Use private or restricted workspaces when available. Student work must not be made public unless public sharing is specifically identified as part of the activity and the required school or board procedures have been followed.

For generative AI, students must follow the stricter PDSB privacy expectations for AI tools and must not place identifying or sensitive information in prompts, uploads, or generated content.

GUARDIAN & STUDENT ACKNOWLEDGEMENT: Please review the course outline, safety expectations, assessment information, and digital-tool expectations. Before signing, caregivers and students should review the linked Safety Package and SOP. It contains the current safety information and describes the tools, machines, equipment, and practical activities the learner may access after completing the required instruction, training, and authorization. Please contact the teacher before signing if you have questions or concerns about this access. Signatures acknowledge that this information has been reviewed.

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| Signature Date | Caregiver/Guardian Signature |
| _____ | _____ |
| Student Name and Student Number | Student Signature |